WASP-36b

R-0848

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_230214` · created 2026-07-28T20:17:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459989.7617 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.69 +/- 1.59 [%]
Orbital Inclination (inc)	87.98 +/- 2.84
Airmass coefficient 1 (a1)	5.98 +/- 3.43
Airmass coefficient 2 (a2)	-1.4 +/- 0.97
Scatter in the residuals of the lightcurve fit is	53.2 %
Best Comparison Star	#1 - [558, 465]
Optimal Aperture	1.11
Optimal Annulus	7.5
Transit Duration (day)	0.093 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1368 (deviation 0.57σ)
Duration fit vs published	0.093 d vs 0.0758 d (deviation 2.12σ)
Mid-time O–C	-3.3 min
Depth SNR	3.4
Residual scatter	53.2 %

Light curve

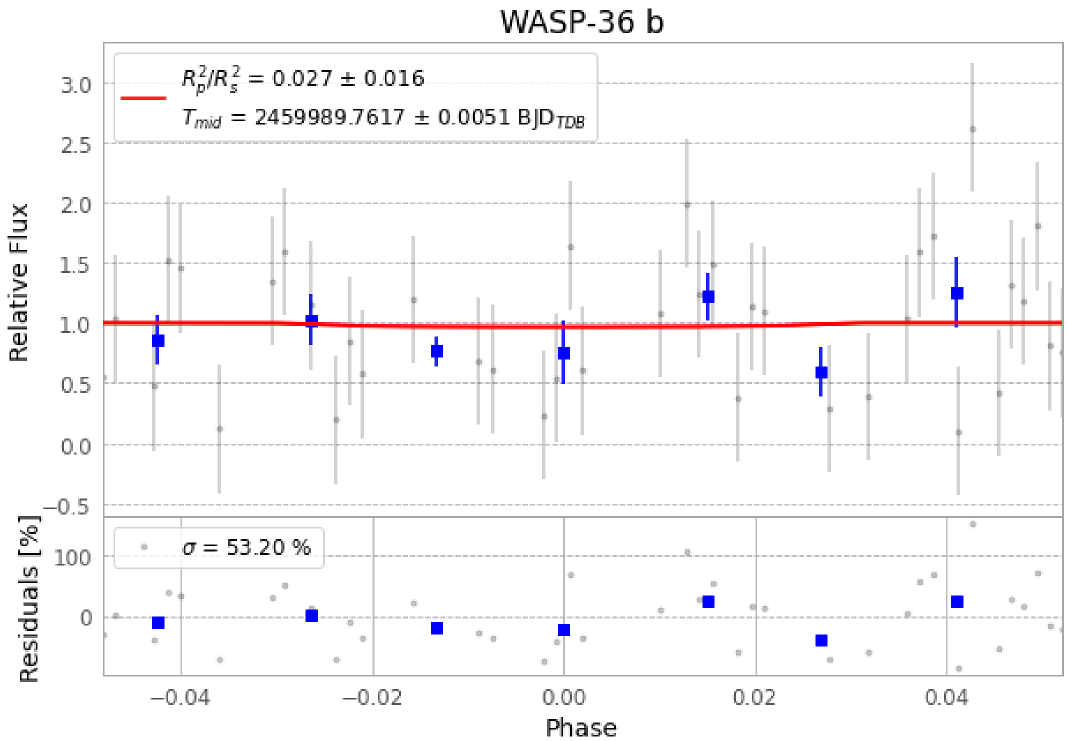

Figure 1 — FinalLightCurve_WASP-36 b_2023-02-13.png (SHA-256 9a153dfc537d937c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [523, 305], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3416380be68f9c81…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-36230214041816.FITS` → `WASP-36230214080615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230214.log (bc8b30daa370…), FinalLightCurve_WASP-36 b_2023-02-13.png (9a153dfc537d…), FinalLightCurve_WASP-36 b_2023-02-13.csv (4cea1c74f512…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 20:17Z.